

Natural Language Processing in Cancer: Extracting Diagnostic Insights from Pathology Reports

Using Traditional NLP · SciBERT · BioBERT · Ensemble
· ChatGPT

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Problem & Motivation

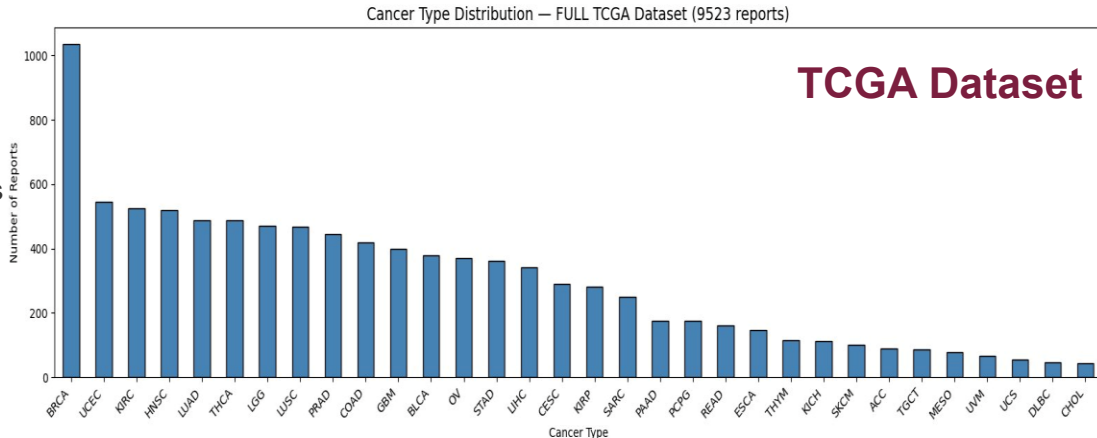
The Problem

Pathology reports are long, complex, unstructured clinical text.

The Cancer Genome Atlas (TCGA) contains 9,523 reports and manual review is impossible.

The Goal

Auto-classify 32 cancer types using NLP from simple word-count models to BERT transformers and ChatGPT.



Why This Matters

Faster Diagnosis

Saves clinicians hours of manual report reading

Research at Scale

Process thousands of reports instantly

Treatment Support

Accurate cancer type drives treatment selection

9,523

Total Reports

32

Cancer Types

7,618

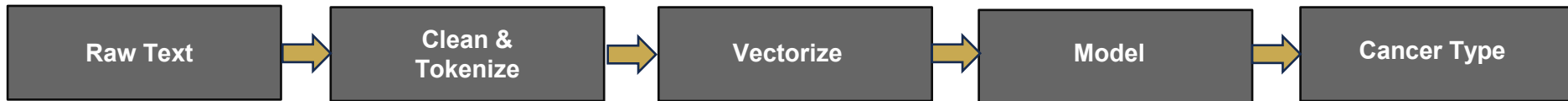
Training

1,905

Testing

Same dataset as Cedars-Sinai AI Campus Project
we exceeded their 89% accuracy → 96.75%

Main Techniques — All 5 Models



Model 1 — Bag of Words + Logistic Regression

- CountVectorizer: top 5,000-word features
- Each report - sparse word-count vector
- Fast, interpretable baseline model
- No word order, word presence/frequency only

Model 2 — TF-IDF + Random Forest

- TF-IDF: down-weights common words, boosts rare/important
- Random Forest Classifier: 100 decision trees, majority vote
- More robust to noisy features than single classifier
- Handles sparse, high-dimensional text data well

Model 3 — SciBERT

- Pretrained on Semantic Scholar scientific papers
- 110M parameters . ALL layers unfrozen
- 10 epochs · T4 GPU · fp16 mixed precision
- Bidirectional encoder, reads full sentence both directions

Model 4 — BioBERT

- Pretrained on PubMed abstracts + PMC full-text
- 108M parameters . ALL layers unfrozen
- 10 epochs · T4 GPU · fp16 mixed precision

Model 5 — Ensemble

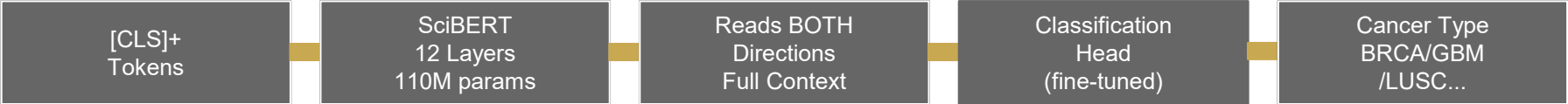
- Combines weighted probabilities from all 3 models
- Leverages strengths of both traditional and transformer
- Best of all models, highest overall accuracy
- Weighted average: TF-IDF 20% + SciBERT 40% + BioBERT 40%

Model 5 — Chat GPT

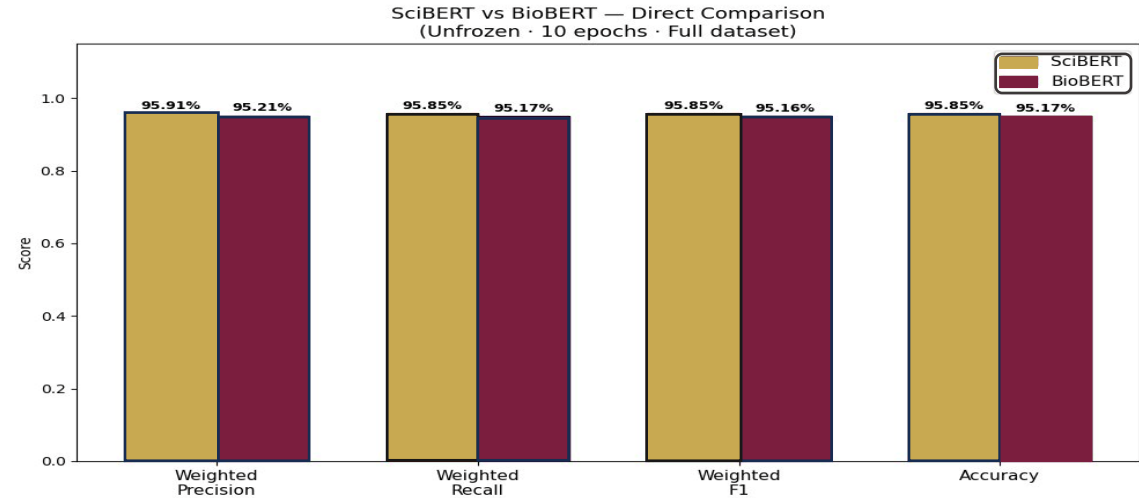
- ChatGPT (GPT-3.5 Turbo): Zero training needed
- prompt-based extraction of cancer type + stage + summary
- completely different from BERT

BERT Transformer vs ChatGPT — How They Work Differently

BERT (SciBERT) — Bidirectional Encoder



Real SciBERT vs BioBERT Comparison



SciBERT WINS: 95.85% vs BioBERT 95.17% , Scientific pertaining generalizes better on TCGA pathology text

ChatGPT — Autoregressive Decoder

Feature	BERT (SciBERT)	ChatGPT
Architecture	Encoder-only	Decoder-only
Reading	Bidirectional	Left → Right
Task	Classify label	Generate text
Training	Fine-tuned on TCGA	Zero-shot prompt
Output	BRCA / GBM / LUSC	Type+Stage+S ummary
Accuracy	95.85% (measured)	Qualitative only

Implementation

1

Load Full Dataset — The Cancer Genome Atlas (TCGA)

9,523 pathology reports · 32 cancer types · Merge reports CSV with cancer label CSV on patient_id

2

Text Preprocessing

Lowercase · Remove digits & punctuation · NLTK stopwords removal · apply clean_text() to all reports

3

Label Encoding + Train / Test Split

32 cancer types → integers 0–31 via label2id · Stratified 80/20 split → 7,618 train / 1,905 test

4

Train 5 Models

BoW+LR · TF-IDF+RF · SciBERT (10ep, ALL unfrozen, fp16, T4 GPU) · BioBERT (10ep, ALL unfrozen, fp16, T4 GPU) · Ensemble

5

Predict + Explain + ChatGPT Extract

predict() function · generate_explanation() with all 32 TCGA cancer types · ChatGPT API prompt-based extraction

10 epochs · ALL 110M BERT parameters unfrozen · Full TCGA dataset

Evaluation & Results

Ensemble Confusion Matrix — Best Model (96.75%)



Actual Classification Report

Ensemble Results (TF-IDF + SciBERT + BioBERT)				
	precision	recall	f1-score	support
ACC	1.00	1.00	1.00	18
BLCA	1.00	0.99	0.99	76
BRCA	1.00	1.00	1.00	207
CESC	0.98	0.97	0.97	58
CHOL	1.00	1.00	1.00	9
COAD	0.95	0.98	0.96	84
DLBC	0.90	1.00	0.95	9
ESCA	0.96	0.93	0.95	29
GBM	0.96	0.99	0.98	80
HNSC	1.00	0.97	0.99	104
KICH	0.95	0.95	0.95	22
KIRC	0.96	0.96	0.96	105
KIRP	0.94	0.91	0.93	56
LGG	0.99	0.97	0.98	94
LIHC	0.96	1.00	0.98	68
LUAD	0.88	0.92	0.90	98
LUSC	0.91	0.87	0.89	94
MESO	1.00	1.00	1.00	16
OV	0.96	0.96	0.96	74
PAAD	0.97	0.94	0.96	35
PCPG	0.97	1.00	0.99	35
PRAD	1.00	1.00	1.00	89
READ	0.97	0.88	0.92	32
SARC	1.00	0.96	0.98	50
SKCM	0.95	1.00	0.98	20
STAD	0.96	0.97	0.97	72
TGCT	1.00	1.00	1.00	17
THCA	1.00	1.00	1.00	98
THYM	1.00	1.00	1.00	23
UCEC	0.92	0.90	0.95	109
UCS	1.00	0.73	0.84	11
UVM	1.00	1.00	1.00	13
accuracy			0.97	1905
macro avg	0.97	0.96	0.97	1905
weighted avg	0.97	0.97	0.97	1905

Results Summary

96.38%
BoW + LR

94.96%
TF-IDF + RF

95.85%
SciBERT

95.17%
BioBERT

96.75%
Ensemble

Conclusion & Future Work

Key Takeaways

- Traditional NLP: BoW 96.38%, TF-IDF 94.96%, medical vocabulary is highly distinctive
- SciBERT WINS vs BioBERT: 95.85% vs 95.17% scientific pretraining generalizes better on TCGA
- Ensemble (TF-IDF + SciBERT + BioBERT) = 96.75% best of all models
- ChatGPT: zero training, generates cancer type + stage + summary completely different from BERT encoder
- Exceeded Cedars-Sinai reference: 89% → 96.75% (+7.75% improvement)

Future Work

- TNM Stage classification -Stage I, II, III, IV prediction
- spaCy named entity recognition for clinical terms
- Hyperparameter tuning via grid search / cross-validation
- Deploy as clinical web app for real-time diagnosis support
- Train on full TCGA clinical endpoint and survival data

9,523

TCGA Reports

32

Cancer Types

5

Models

96.75%

Best Accuracy

0

ChatGPT Training

Thank you!